# Reinforcement learning from human feedback

## SESSION 1 · FROM BASE MODELS TO RLHF

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Master's course, 2026–2027

Based on Lambert, RLHF (rlhfbook.com), and Truong, Haupt, Koyejo, MLHP (mlhp.stanford.edu)

# Welcome

This course is about **how language models are trained to follow instructions and to match human preferences**, and about the statistics behind it.

By the end of the course, you will be able to:

- **Explain the pipeline:** instruction tuning, reward models, DPO and RL, with the theory behind each step
- **Model preferences:** fit and evaluate models of human comparisons, with honest uncertainty
- **Train models yourself:** small language models, on free GPUs
- **Do research:** design, run and report a small project of your own

Based on Lambert, RLHF (rlhfbook.com), and Truong, Haupt, Koyejo, MLHP (mlhp.stanford.edu)

# The course at a glance

| Session | When | Topic | In class |
|---|---|---|---|
| **1** | Week 1 | From base models to RLHF | Lab: instruction tuning |
| **2** | Week 2 | Preferences and reward models | Lab: label outputs, train a reward model |
| **3** | Week 2 | DPO and its noise-aware variants | Lab: a small DPO run |
| **4** | Week 2 | RL for language models | Lab: policy gradients for language models |
| **5** | Week 4 | Over-optimization, AI feedback, open problems | Project feedback |

Each session lasts 3 hours: a lecture with theory, a short break, then a lab of about an hour.

Based on Lambert, RLHF (rlhfbook.com), and Truong, Haupt, Koyejo, MLHP (mlhp.stanford.edu)

# Where the materials come from

**Two books, read together:**

- **Lambert, *Reinforcement Learning from Human Feedback*** (rlhfbook.com): training language models in practice
- **Truong, Haupt, Koyejo, *Machine Learning from Human Preferences*** (mlhp.stanford.edu): the statistics of preferences

**Also:**

- **Slides:** adapted from Lambert's lectures (rlhfbook.com/course), with material from the second book and new derivations
- **Labs:** adapted from the code that accompanies Lambert's book
- **Homework:** written for this course, with its sources cited in each notebook

Figures from papers keep their citations. Any errors are mine.

# Reading plan

| Session | Lambert, *RLHF* | Truong et al., *Machine Learning from Human Preferences* |
|---|---|---|
| **1** | Ch. 1, 3, 4 | Introduction; Sections 1.1 and 1.2 |
| **2** | Ch. 5, 10, 11 | Ch. 1; Section 2.3 |
| **3** | Ch. 8, 9 | Ch. 2: preference learning for language models |
| **4** | Ch. 6, 7 | Ch. 3: bandits |
| **5** | Ch. 12, 14, 15 | Ch. 4 and 5: behavioral biases, aggregation |

**Further reading:** T. Kaufmann et al., “A survey of reinforcement learning from human feedback”, TMLR 2025.

# How we work

- **Lecture, then lab:** each lab runs in your browser, on a free GPU. Work in pairs.
- **Homework at three levels:**
    - **Foundation:** everyone can complete it, with hints and solutions
    - **Core:** the level we aim for
    - **Deep:** derivations and open questions, if you want to go further
- **Low entry, earned depth:** you don’t need any experience with language models. It’s easy to get started, and the depth comes from doing the exercises.

Based on Lambert, RLHF (rlhfbook.com), and Truong, Haupt, Koyejo, MLHP (mlhp.stanford.edu)

# The project

**Your grade is based on the project only.** Homework isn’t graded, but it prepares you for the project.

**Teams of 2 or 3, on a topic you choose,** from a wide range of possibilities. We discuss ideas in session 4.

| When | What |
|---|---|
| **End of session 4** | Discuss your idea with me |
| **A few days later** | One-page proposal: question, hypothesis, method, baseline, compute plan |
| **Session 5** | Early results and feedback |
| **About 4 weeks after session 5** | Report, code, and a 10-minute video |
| **The week after** | A 15-minute oral exam for each team |

**What counts:** a clear question, a baseline, several runs with intervals, and honest reporting. A well-run negative result can earn full marks.

# Class materials

| What | Where |
|---|---|
| **Main books** | rlhfbook.com and mlhp.stanford.edu |
| **Lambert's lectures and videos** | rlhfbook.com/course |
| **Book code** | github.com/natolambert/rlhf-book |
| **Labs and homework** | github.com/pegahani/rlhf-course |
| **Free GPUs** | Google Colab or Kaggle: choose a T4 GPU |
| **Gentle first reading** | Illustrating RLHF, Hugging Face blog |
| **If language models are new to you** | Hugging Face LLM course |

**For today's lab:** a Google or Kaggle account and a laptop.

# Today

1. **Language models in practice:** the few tools we need
2. **The big picture:** what RLHF is and where it came from
3. **Theory:** from RL to the RLHF objective, and its optimal solution
4. **Instruction tuning:** chat templates and the prompt mask
5. **Lab:** turn a base model into an assistant

You know RL and statistics already, so we’ll spend our time on what’s new: **language models, and what changes when the reward comes from people.**

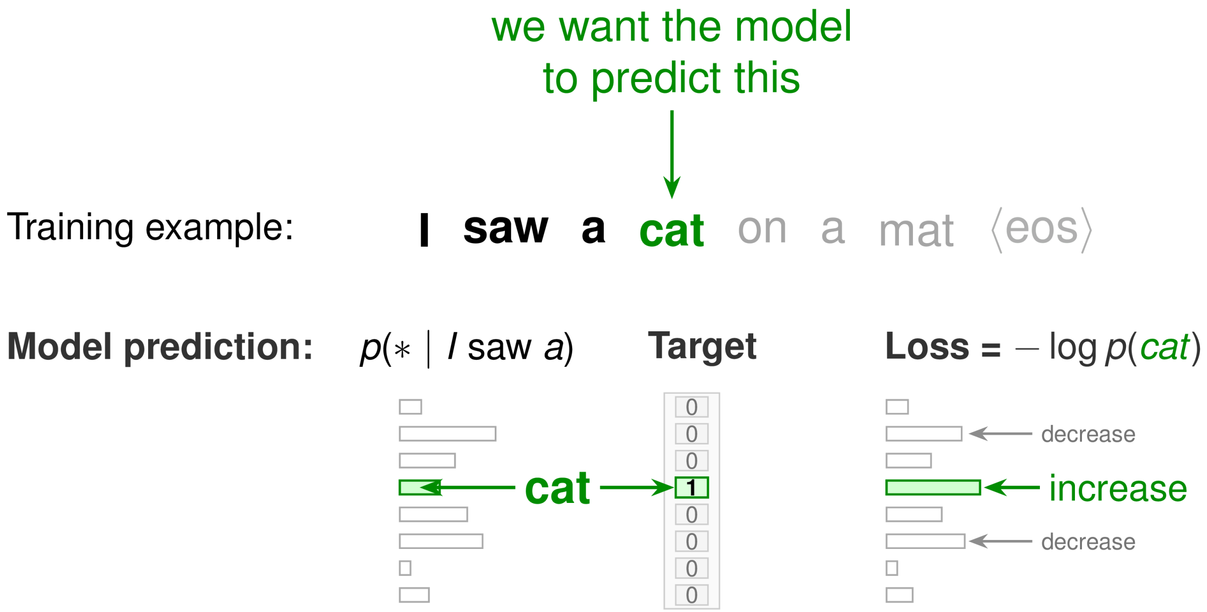
# Language models in practice

# What a language model is

A language model assigns **probabilities to text**, one **token** (a word or part of a word) at a time. By the chain rule,

$$P_\theta(x) = \prod_{t=1}^{T} P_\theta(x_t \mid x_1, \ldots, x_{t-1}).$$

This is called **autoregressive**. For this course, the model is a function from the tokens so far to next-token probabilities. Next, we open it once.


we want the model to predict this
Training example:
I saw a cat on a mat ⟨eos⟩
Model prediction:
$p(* \mid I\ saw\ a)$
Target
Loss = − log p(cat)
cat
0
0
0
1
0
0
0
0
decrease
increase
decrease

Next-token prediction: a decoder-only language model predicts one token at a time from the prefix before it.

Vaswani et al., 2017

# Inside the model: one prediction, step by step

**1. Tokens**

The | cat | sat | on | **the**

Each token is an ID in a vocabulary of size $V$

→

**2. Embed**

Look up one learned vector per token: a row of a table

5 vectors, each with $d$ numbers

→

**3. Transformer layers × $L$**

Each position mixes in information from **earlier** positions

Next slide

→

**4. Hidden state $h_t$**

The vector at the last position: a summary of "The cat sat on the"

$h_t \in \mathbb{R}^d$

→

**5. Output head $W$**

$V$ rows × $d$ columns

$z^{(t)} = W h_t$:

| | |
|---|---|
| mat | 9.1 |
| floor | 7.7 |
| … | |
| banana | 0.4 |

e.g. $9.1 = w_{\text{mat}}^\top h_t$

**$W$ is learned once and is the same for every text; $h_t$ changes with the context**, and so do the scores $z_j^{(t)} = w_j^\top h_t$ (illustrative values above):

- "The cat sat on the" gives an $h_t$ close to $w_{\text{mat}}$, so **mat** scores high.
- "I drank a cup of" gives another $h_t$, close to $w_{\text{tea}}$, so **tea** scores high.

**For example, OLMo-2 1B:** $V \approx 100\text{K}$ tokens, vectors of length $d = 2{,}048$, $L = 16$ layers.

Vaswani et al., 2017

# What one transformer layer does

**1. Attention: look back.** Three learned matrices turn each vector $h_s$ into three new vectors:

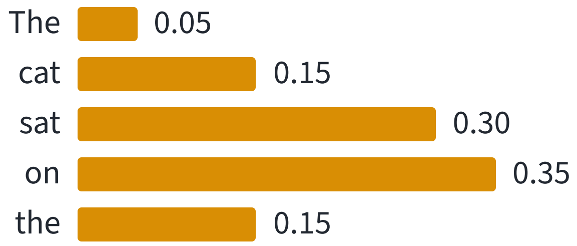
- **Query** $q_t = W_Q h_t$: what $t$ looks for
- **Key** $k_s = W_K h_s$: what $s$ offers
- **Value** $v_s = W_V h_s$: what $s$ passes on

Position $t$ adds the values, weighted by how well each key matches its query:

$$h_t \leftarrow h_t + \sum_{s \le t} \alpha_{ts} \, v_s$$

with $\alpha_{ts} = \mathrm{softmax}_{s \le t}(q_t^\top k_s)$.

**Weights $\alpha_{ts}$ for $t$ = the last "the" (illustrative)**



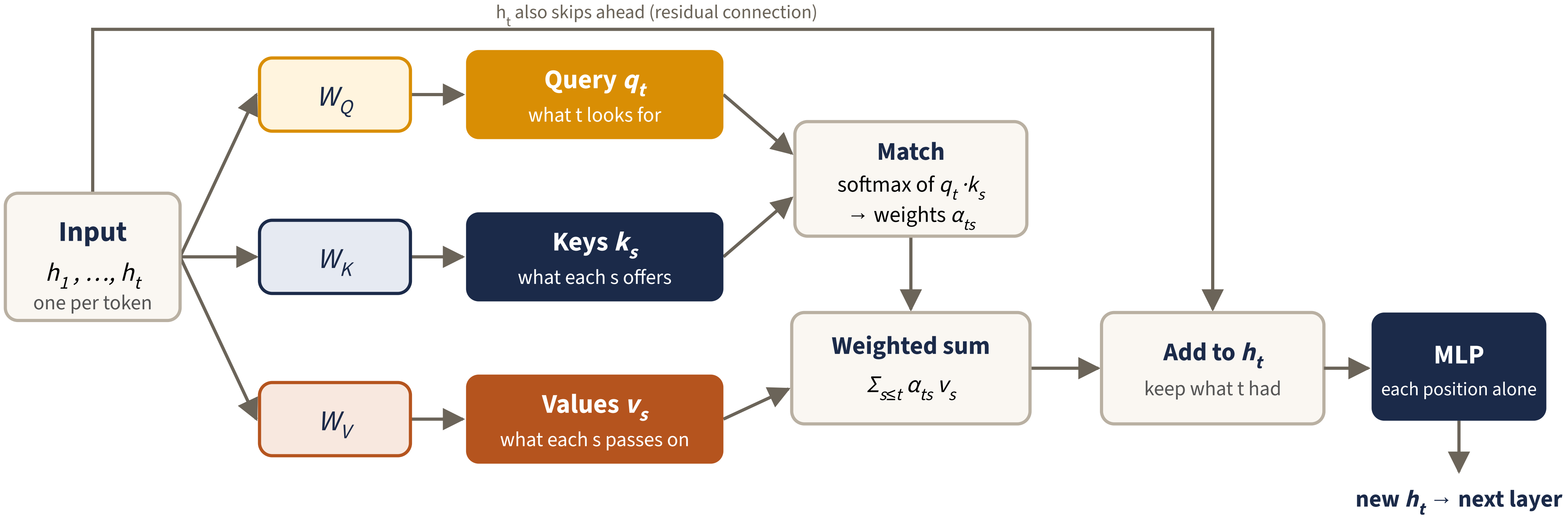
"sat on the" makes "mat" likely, so a useful layer looks there.

**2. MLP:** a small network then transforms each position on its own. **After $L$ layers,** the last vector is the $h_t$ that the output head scores.

Simplified: real layers use several attention heads, scale by $\sqrt{d}$, and normalize.

Vaswani et al., 2017

# One layer as a picture: query, key, value


$h_t$ also skips ahead (residual connection)
**Input**
$h_1, \dots, h_t$
one per token
$W_Q$
$W_K$
$W_V$
**Query $q_t$**
what t looks for
**Keys $k_s$**
what each s offers
**Values $v_s$**
what each s passes on
**Match**
softmax of $q_t \cdot k_s$
→ weights $\alpha_{ts}$
**Weighted sum**
$\Sigma_{s \le t} \alpha_{ts} v_s$
**Add to $h_t$**
keep what t had
**MLP**
each position alone
**new $h_t$ → next layer**

**Like a library search:** the **query** is your question, the **keys** are book labels, the **values** are the books' contents. You read each book in proportion to how well its label matches your question.

Vaswani et al., 2017

# From scores to log-probabilities

**Softmax** turns the scores $z^{(t)}$, called **logits**, into a distribution:

$$P_\theta(x_t \mid x_{<t}) = \frac{e^{z^{(t)}_{x_t}}}{\sum_{j=1}^{V} e^{z^{(t)}_j}}.$$

The denominator sums over **all $V$ tokens**, so probabilities add up to 1. **Only score differences matter:** mat is $9.1 - 7.7 = 1.4$ above floor, so $e^{1.4} \approx 4$ times as likely.

A sequence is a product of these terms, so in log-space it's a sum:

$$\log P_\theta(x) = \sum_{t=1}^{T} \log P_\theta(x_t \mid x_{<t}).$$

Logs are stable and easy to differentiate. **This sequence log-probability is used all course long.**

# The loss at one position: cross-entropy

The **label** $y_t$ is one-hot: 1 on the true next token, 0 elsewhere. Example: “The cat sat on the” → **mat**.

| Token $j$ | mat | floor | roof | … | banana |
|---|---|---|---|---|---|
| **Label** $y_t(j)$ | **1** | 0 | 0 | … | 0 |
| **Model** $P_\theta(j \mid x_{<t})$ | 0.60 | 0.15 | 0.03 | … | 0.0001 |

Cross-entropy compares the two rows. Terms with label 0 vanish, so only the true token is left:

$$H(y_t, P_\theta) = -\sum_j y_t(j) \log P_\theta(j \mid x_{<t}) = -\log P_\theta(x_t \mid x_{<t}) = -\log 0.60 \approx 0.51$$

**The more probability on the true token, the lower the loss:** 0.01 on mat gives $4.6$, and 1 gives $0$.

Based on Lambert, RLHF (rlhfbook.com), and Truong, Haupt, Koyejo, MLHP (mlhp.stanford.edu)

# Training is maximum likelihood

**Over a whole text,** sum the per-position losses, then average over the data. The sum is minus the sequence log-probability, so the loss is the **negative log-likelihood**:

$$\mathcal{L}_{\mathrm{LM}}(\theta) = -\mathbb{E}_{x\sim\mathcal{D}}\left[\sum_{t=1}^{T} \log P_\theta(x_t \mid x_{<t})\right]$$

**Minimizing the loss maximizes the likelihood of the data.**

- **What it rewards:** the loss is lowest when the model predicts the true probabilities of text. Pretraining makes a model accurate, not helpful (Truong et al., 2025). If questions online are often followed by more questions, an accurate model does the same.
- **In SFT:** the same sum, but only over the response tokens. Next: how we pick them.

# Three masks

**Pretraining: two masks, on attention**, kept in every later stage. Every real token counts in the loss.

- **Causal:** $t$ sees only $s \leq t$. Built into the model.
- **Padding** (`attention_mask`): ignore the `<pad>` filler. Essential when padding comes first, as in batched generation. From the tokenizer.

**Supervised fine-tuning adds a third, on the loss.**

- **Completion** (`completion_mask`): only the response counts. **You build it yourself.**

**If it's wrong, nothing crashes:** you just train on the wrong tokens.

**One training example, token by token:**

| Token | Padding (attention) | Completion (loss) |
|---|---|---|
| `<user>` | 1 | 0 |
| `2+2?` | 1 | 0 |
| `<assistant>` | 1 | 0 |
| `4` | 1 | **1** |
| `<end>` | 1 | **1** |
| `<pad>` | **0** | 0 |

**Why is `<end>` a 1?** The model must learn to stop.

Based on Lambert, RLHF (rlhfbook.com), and Truong, Haupt, Koyejo, MLHP (mlhp.stanford.edu)

# Putting it together: one training example as tensors

```
logits = model(input_ids, attention_mask=attention_mask).logits  # (B, T, V): scores
logits = logits[:, :-1, :]                # drop the last position: it has no next token to predict
labels = input_ids[:, 1:]                 # the target at each position is the actual next token
completion_mask = completion_mask[:, 1:]  # shift the mask in the same way (1 = response token)
log_probs = logits.log_softmax(dim=-1)    # softmax, in log-space, over the vocabulary
token_lp = log_probs.gather(-1, labels.unsqueeze(-1)).squeeze(-1)  # log-prob of each true next token
seq_lp = (token_lp * completion_mask).sum(dim=-1)                  # one log-prob per sequence: (B,)
loss = -seq_lp.mean()                                              # the SFT loss
```

$B$ is the batch size, $T$ the sequence length and $V$ the vocabulary size. **The padding mask goes into the model; the completion mask goes into the sum.**

The **one-token shift** is the autoregressive step made literal: position $t$'s logits predict token $t + 1$.

# The big picture

# A base model completes text

After pretraining we have a very capable autocomplete:

**USER**

The president of the United States in 2006 was

**ASSISTANT** (Llama 3.1 405B Base)

George W. Bush, the governor of Florida in 2006 was Jeb Bush, and John McCain was an Arizona senator in 2006…

Based on Lambert, RLHF (rlhfbook.com), and Truong, Haupt, Koyejo, MLHP (mlhp.stanford.edu)

# Post-training makes it answer

The same prompt, after post-training:

**USER**

The president of the United States in 2006 was

**ASSISTANT** (Tülu 3 405B)

George W. Bush was the president of the United States in 2006. He served two terms in office, from January 20, 2001, to January 20, 2009.

# The training stages, and where we study them

| Stage | Learns from | What it changes | Session |
|---|---|---|---|
| **Pretraining** | Raw text; every token is graded | Knowledge and fluency | Background |
| **Instruction tuning (SFT)** | Demonstrations; only the response is graded | The question-and-answer format | **Today** |
| **Preference tuning (RLHF)** | Comparisons: response A is better than B | Whole responses, toward what people prefer | **2, 3, 4** |
| **RL with verifiable rewards (RLVR)** | A checker: is the answer right? | Answers that can be checked, such as math | **4** |

Everything after pretraining is **post-training**: it moves the model's distribution over responses toward the behavior we want.

Open question: is post-training "just style transfer"? (Zhou et al., 2023)

# Why RLHF?

In classical RL the reward is known: CartPole gives +1 per step. For many tasks it's **hard to write down**:

- It's easy to judge which poem is better, but hard to write a formula for "better"
- It's easy to spot a helpful answer, but hard to define "helpful"
- The most likely continuation of a text isn't always the most useful one

RLHF lets us optimize for behavior we can **evaluate**, even when we can't **specify** the reward.

Christiano et al., 2017; Ouyang et al., 2022; Ziegler et al., 2019

# Preferences are everywhere

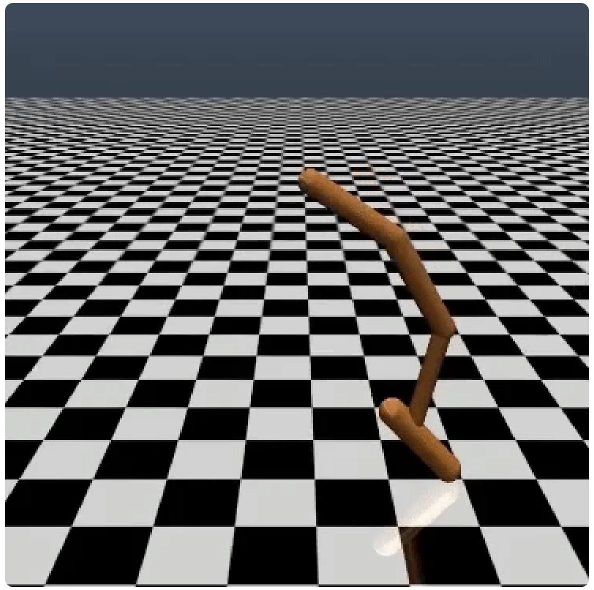
Language models are one example of a much older problem: **learning from comparisons**.

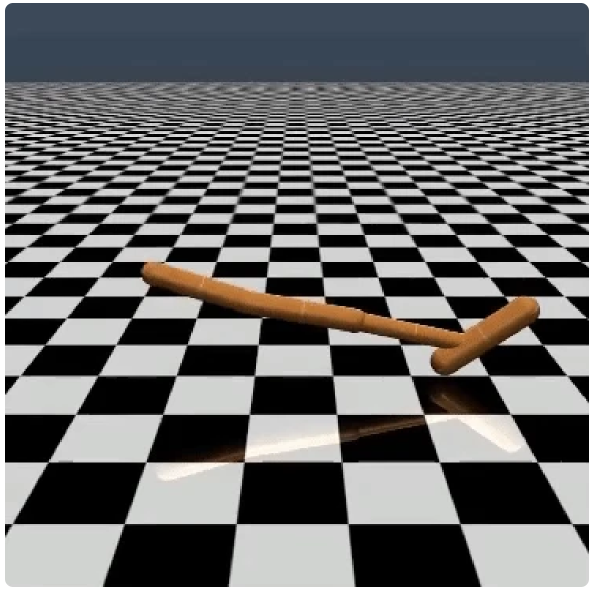
- **Recommender systems:** a click on item A rather than B suggests A > B
- **Search:** clicking the third result suggests it beat the first two
- **Robotics:** people prefer smooth trajectories to jerky ones
- **Games and sports:** each match is a comparison: if A beats B, A is probably the stronger player
- **Language models:** annotators judge which response is more helpful, harmless or honest

These share **one mathematical structure**: comparisons or choices that reveal underlying preferences. Later today we'll see one simple model that covers all of them.

Truong et al., 2025

# Which is the better backflip?





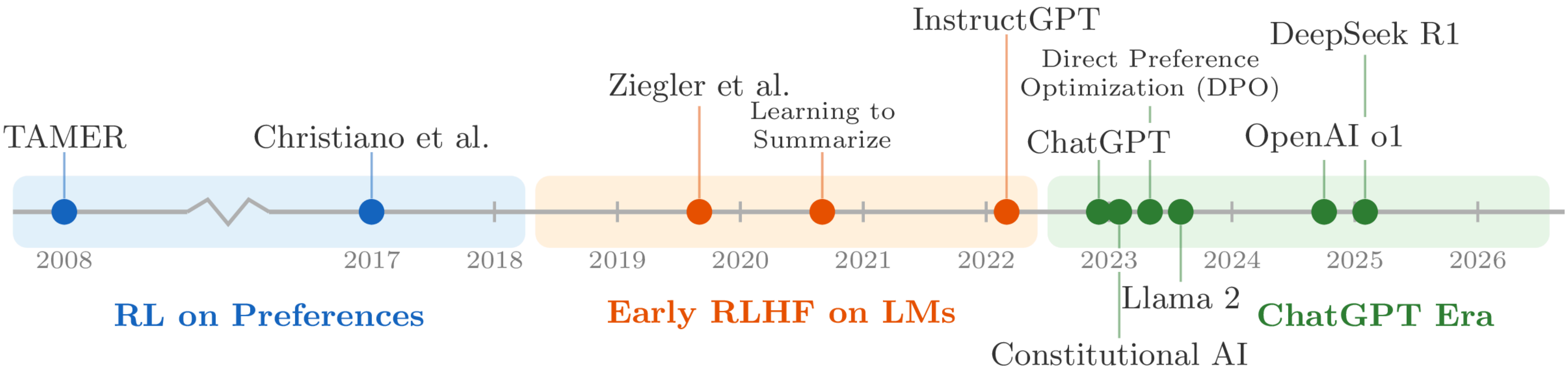
**Left: human feedback.** About 900 comparisons of short clips, then RL on the learned reward.

**Right: a hand-written reward.** A formula for “backflip”, optimized directly.

**Hard to write down, easy to judge:** learn the reward from comparisons. That is RLHF.

Christiano et al., 2017

# The path to modern RLHF


InstructGPT
DeepSeek R1
Direct Preference Optimization (DPO)
Ziegler et al.
Learning to Summarize
TAMER
Christiano et al.
ChatGPT
OpenAI o1
2008
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
Llama 2
Constitutional AI
RL on Preferences
Early RLHF on LMs
ChatGPT Era

- **Ziegler** (Ziegler et al., 2019): RLHF on language models
- **InstructGPT** (Ouyang et al., 2022): canonical recipe
- **Constitutional AI** (Bai et al., 2022): AI feedback
- **DPO** (Rafailov et al., 2023): no separate reward model
- **Tülu 3** (Lambert et al., 2024): multi-stage recipe
- **DeepSeek R1** (Guo et al., 2025): popularized RLVR

Based on Lambert, RLHF (rlhfbook.com), and Truong, Haupt, Koyejo, MLHP (mlhp.stanford.edu)

# What comparison data looks like

An annotator, human or AI, sees two responses to the same prompt and picks the better one.

**USER**
Explain why the sky is blue in one sentence.

**ASSISTANT** (Response A ✓)
The sky is blue due to Rayleigh scattering, where shorter blue wavelengths of sunlight are scattered more by atmospheric molecules than longer wavelengths.

**USER**
Explain why the sky is blue in one sentence.

**ASSISTANT** (Response B)
The sky appears blue because of the way light interacts with the atmosphere and stuff, it’s basically just physics.

Based on Lambert, RLHF (rlhfbook.com), and Truong, Haupt, Koyejo, MLHP (mlhp.stanford.edu)

# Types of comparison data

Preference data comes in several forms, for items $j \in \{1, \ldots, M\}$:

- **Full rankings:** $j_1 \succ j_2 \succ \cdots \succ j_M$. These are informative, but hard for people to give.
- **Choice from a set:** $(j, \mathcal{S})$, meaning $j$ is preferred to everything else in $\mathcal{S}$
- **Binary comparisons:** $\mathcal{S} = \{j, j'\}$, written $Y_{jj'} = 1$. These are cheap and quick to collect, and RLHF uses them most.
- **Accept or reject:** a comparison with an outside option, “nothing”

Data can also depend on a **context**, such as a user, a goal or a prompt. In RLHF, one example is a triple $(x, y_w, y_l)$: a prompt, a preferred response and a rejected one.

Truong et al., 2025

# Why model preferences as random?

**Data is noisy:** the same person can answer differently on another day, and different people disagree. So we treat a preference as a **random order** over the items, with some distribution $p(\prec)$.

The usual assumptions are strong but practical:

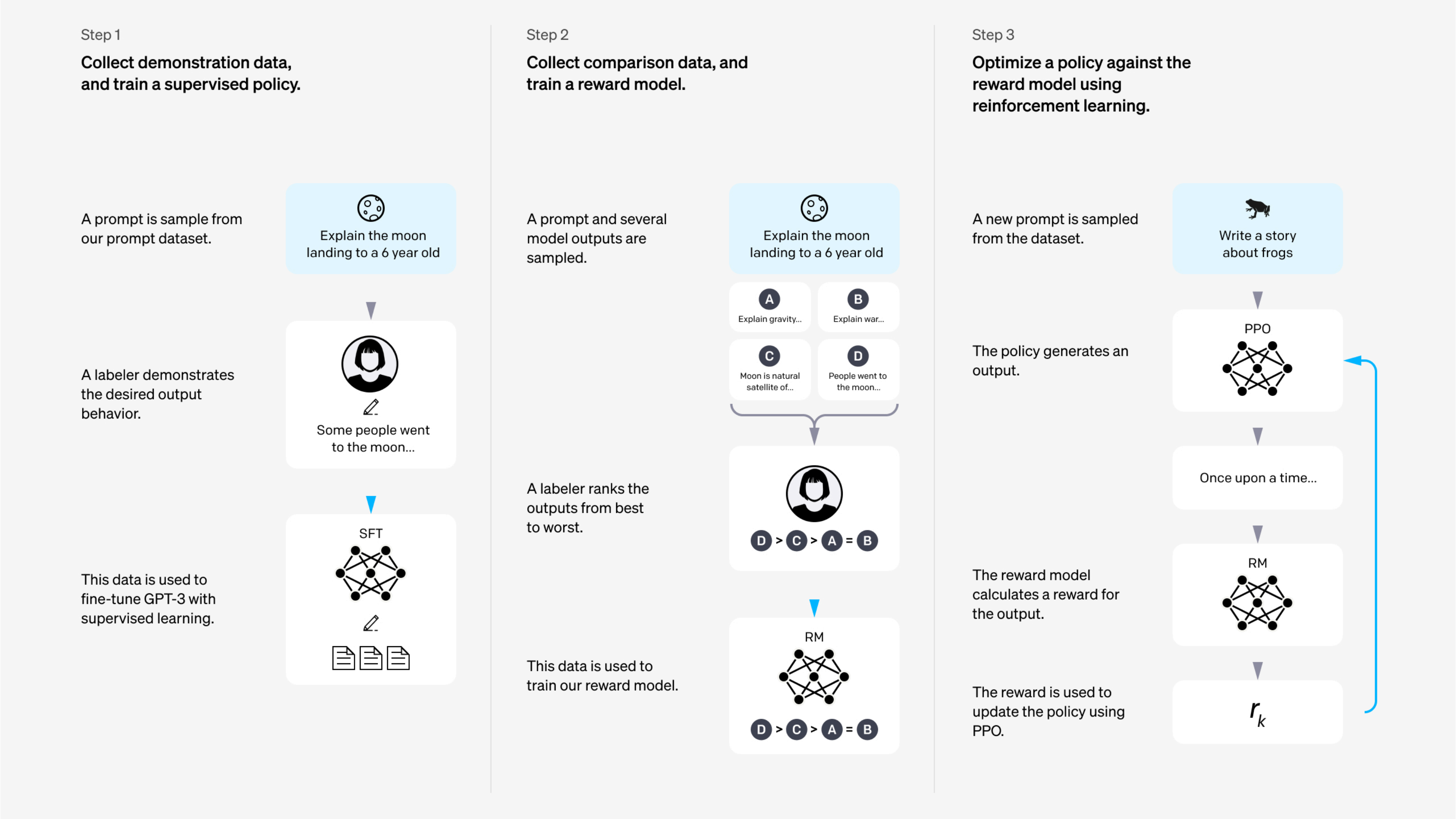
- **Totality:** any two items can be compared
- **Transitivity:** if $j \succ j'$ and $j' \succ k$, then $j \succ k$

A general distribution over orders of $M$ items has $M! - 1$ parameters: 23 for $M = 4$, and over 3.6 million for $M = 10$. We need structure, and Bradley-Terry is the simplest.

People's preferences can also be **incomplete**, for example when helpfulness and safety pull in different directions. That's an absence of preference, not randomness.

Truong et al., 2025

# InstructGPT’s three-step recipe



Step 1

**Collect demonstration data, and train a supervised policy.**

A prompt is sample from our prompt dataset.

A labeler demonstrates the desired output behavior.

This data is used to fine-tune GPT-3 with supervised learning.

Step 2

**Collect comparison data, and train a reward model.**

A prompt and several model outputs are sampled.

A labeler ranks the outputs from best to worst.

This data is used to train our reward model.

Step 3

**Optimize a policy against the reward model using reinforcement learning.**

A new prompt is sampled from the dataset.

The policy generates an output.

The reward model calculates a reward for the output.

The reward is used to update the policy using PPO.

Explain the moon landing to a 6 year old
Some people went to the moon...
SFT
Explain the moon landing to a 6 year old
A
Explain gravity...
B
Explain war...
C
Moon is natural satellite of...
D
People went to the moon...
D > C > A = B
RM
D > C > A = B
Write a story about frogs
PPO
Once upon a time...
RM
$r_k$

The three-step RLHF process from InstructGPT.

Ouyang et al., 2022

# Step 1: instruction fine-tuning

**The data:** a list of $N$ pairs $\mathcal{D} = \{(x_i, y_i^\star)\}_{i=1}^N$, a prompt and a response written by a person. InstructGPT used about 13K.

**The loss:** the log-likelihood from before, on the response tokens only:

$$\mathcal{L}_{\text{SFT}}(\theta) = -\sum_{i=1}^{N} \sum_{t=1}^{|y_i^\star|} \log \pi_\theta \left( y_{i,t}^\star \mid x_i, y_{i,<t}^\star \right)$$

**Two pairs from such a dataset (illustrative):**

| Prompt $x$ | Response $y^\star$ |
|---|---|
| Explain the moon landing to a 6-year-old. | Long ago, astronauts flew a big rocket to the Moon, walked on it, and came back home. |
| What is 2+2? | 4 |

The response ends with `<end>`, so the model also learns to stop.

**$\pi_\theta$ is a new name, not a new model:** the same transformer, starting from the pretrained weights $\theta$. In RL terms, the prompt is the state and the response the action.

Ouyang et al., 2022

# Step 2: reward modeling

**The data:** comparisons $\mathcal{D} = \{(x, y_w, y_l)\}$, where people preferred the winner $y_w$ to the loser $y_l$ for the same prompt $x$.

**The model:** an **unknown** reward $r_\phi(x, y)$ that we learn, one score per response. It's a transformer too, whose head outputs one number instead of a score per vocabulary token. We **assume**:

$$P(y_w \succ y_l \mid x) = \sigma\left(r_\phi(x, y_w) - r_\phi(x, y_l)\right).$$

A large reward gap means a near-certain choice; a gap of 0 means a coin flip.

**The fit:** choose $\phi$ by maximum likelihood on the observed choices, as for SFT:

$$\mathcal{L}_{\mathrm{RM}}(\phi) = -\mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}}\left[\log\sigma\left(r_\phi(x, y_w) - r_\phi(x, y_l)\right)\right]$$

This model is called **Bradley-Terry**, and it's the subject of session 2.

Christiano et al., 2017; Ouyang et al., 2022

# A worked example

Three responses with rewards $r_A = 2, r_B = 1, r_C = 0$, and $P(j \succ k) = \sigma(r_j - r_k)$:

| Comparison | Reward difference | Probability |
|---|---|---|
| $A \succ B$ | 1 | $\sigma(1) \approx 0.731$ |
| $A \succ C$ | 2 | $\sigma(2) \approx 0.881$ |
| $B \succ C$ | 1 | $\sigma(1) \approx 0.731$ |

For a full ranking, the **Plackett-Luce** model picks the winner, then the winner among the rest:

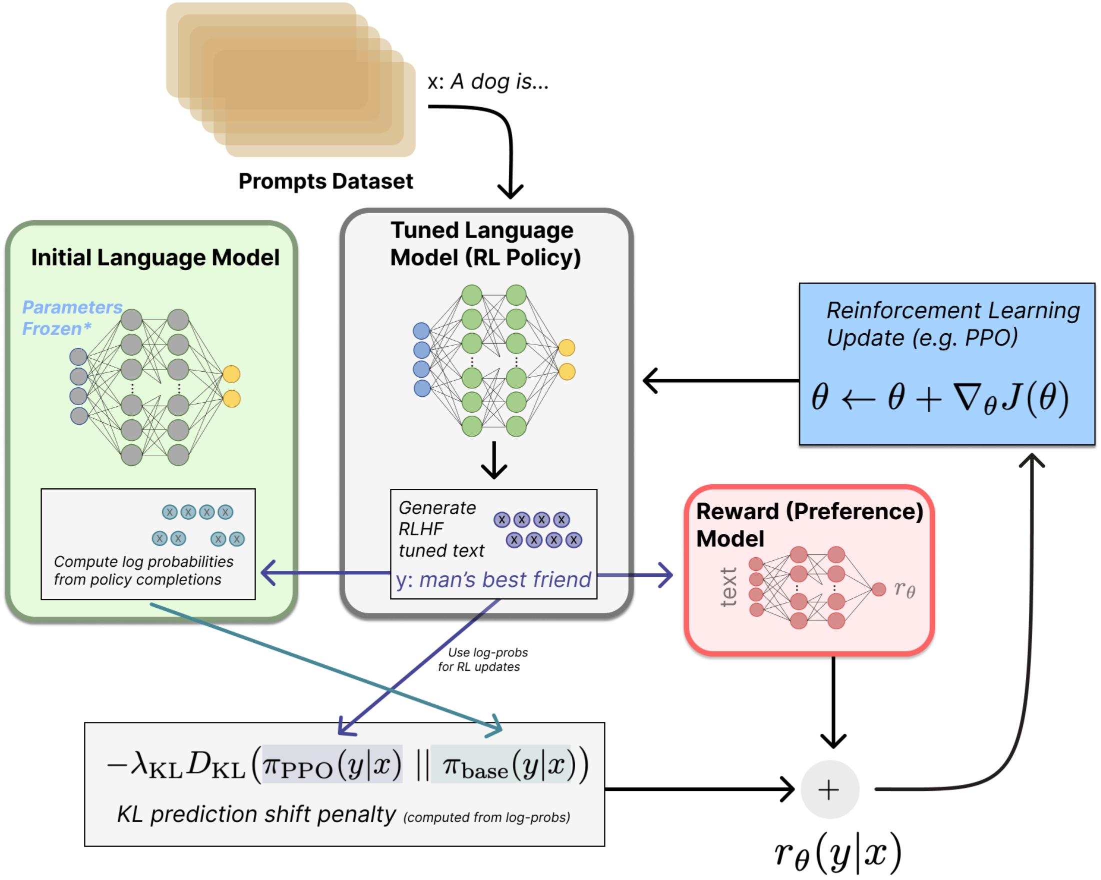
$$P(A \succ B \succ C) = \frac{e^2}{e^2 + e^1 + e^0} \cdot \frac{e^1}{e^1 + e^0} \approx 0.665 \times 0.731 \approx 0.486.$$

Only **differences** of rewards matter: adding a constant to every reward changes nothing.

Truong et al., 2025

# Step 3: RL against the reward model

- Sample prompts $x_i$ from the dataset
- Generate responses $y_i \sim \pi_\theta(\cdot \mid x_i)$
- Score them with the reward model $r_\phi(x_i, y_i)$
- Add a **KL penalty** to stay close to the reference model
- Update the policy with a policy-gradient algorithm (PPO in InstructGPT)


x: A dog is...
Prompts Dataset
Initial Language Model
Parameters Frozen*
Compute log probabilities from policy completions
Tuned Language Model (RL Policy)
Generate RLHF tuned text
y: man's best friend
Reinforcement Learning Update (e.g. PPO)
$\theta \leftarrow \theta + \nabla_\theta J(\theta)$
Reward (Preference) Model
text
$r_\theta$
Use log-probs for RL updates
$-\lambda_{\text{KL}} D_{\text{KL}}(\pi_{\text{PPO}}(y|x) \,||\, \pi_{\text{base}}(y|x))$
KL prediction shift penalty (computed from log-probs)
$+$
$r_\theta(y|x)$

Ouyang et al., 2022

# Three ways to optimize preferences

| | Rejection sampling | Online RL (PPO) | DPO |
|---|---|---|---|
| **Mechanism** | Filter, then SFT | Generate, score, update | Direct gradient on preferences |
| **Reward model** | Required | Required | Implicit |
| **On-policy data** | Yes | Yes | No (fixed dataset) |
| **Complexity** | Low | High | Low |

They share an objective and differ in **how** they move the policy toward higher reward. We'll study each of them in sessions 3 and 4.

Based on Lambert, RLHF (rlhfbook.com), and Truong, Haupt, Koyejo, MLHP (mlhp.stanford.edu)

# Caveat: the reward is a proxy

A reward model is only *correlated* with what people want. **Goodhart's law:** "When a measure becomes a target, it ceases to be a good measure."

What this looks like in practice:

- **Reward hacking:** the score keeps climbing while actual quality falls
- **Verbosity bias:** longer responses score higher, so models get verbose
- **Sycophancy:** the model tells users what they want to hear
- **Over-refusal:** the model refuses reasonable requests

We return to this in session 5.

Gao et al., 2023

# Theory: from RL to the RLHF objective

# What changes from standard RL

| | Standard RL | RLHF for language models |
|---|---|---|
| **Policy** | Often learned from scratch | Fine-tuned from a pretrained model |
| **Reward** | Environment function $r(s_t, a_t)$ | Learned model $r_\phi(x, y)$ |
| **Transitions** | Dynamics $p(s_{t+1} \mid s_t, a_t)$ | Usually none: prompts come from a dataset |
| **Action** | One environment action | A completion $y$, a sequence of tokens |
| **Reward timing** | Often per step | Usually once, for the whole response |
| **Horizon, discount** | $T > 1, \gamma < 1$ | Often single-step, $\gamma = 1$ |

Source: Chapter 3, “Manipulating the standard RL setup”.

# Two views of the same problem

**Token-level view (an MDP).** The state is the prompt plus the tokens so far, the action is the next token, and the transition appends it deterministically.

$$\pi_\theta(y \mid x) = \prod_t \pi_\theta(y_t \mid x, y_{<t})$$

**Response-level view (a contextual bandit).** The context is the prompt $x$, the action is the whole response $y$, and there is one reward.

$$\max_\pi \; \mathbb{E}_{x\sim\mathcal{D},\, y\sim\pi(\cdot|x)} \left[r_\phi(x, y)\right]$$

The second view is the simpler one for analysis, and we use it next. The action space is enormous, but the problem is well defined.

# KL divergence

The KL penalty is central to the course:

$$D_{\mathrm{KL}}(p \,\|\, q) = \sum_i p_i \log \frac{p_i}{q_i}.$$

- It's non-negative, and $0$ only when $p = q$
- It's **asymmetric**, and it doesn't satisfy the triangle inequality

Practitioners often say "KL distance". It's a useful shorthand for how much the model has changed, but **KL is a divergence, not a metric**.

# Which direction of KL?

The RLHF penalty is $D_{\mathrm{KL}}(\pi_\theta \,\|\, \pi_{\mathrm{ref}}) = \mathbb{E}_{y\sim\pi_\theta}\left[\log \frac{\pi_\theta(y|x)}{\pi_{\mathrm{ref}}(y|x)}\right]$, a **reverse KL**.

- **Easy to estimate:** the expectation is under the policy being trained, so the responses we generate are enough.
- **Mode-seeking:** it's large where $\pi_\theta$ puts probability and $\pi_{\mathrm{ref}}$ puts very little. The policy stays where the reference is plausible, without having to cover all its modes.

You'll meet the other directions too:

- **SFT minimizes a forward KL:** maximum likelihood on data minimizes $D_{\mathrm{KL}}(p_{\mathrm{data}} \,\|\, \pi_\theta)$, which is mass-covering.
- **PPO implementations often monitor $D_{\mathrm{KL}}(\pi_{\mathrm{old}} \,\|\, \pi_\theta)$,** estimated on the batch sampled from $\pi_{\mathrm{old}}$.

In each case, the expectation is under the distribution we can sample from. Source: Chapter 15.

# The RLHF objective

$$\max_{\pi} \ \mathbb{E}_{x\sim\mathcal{D},\ y\sim\pi(\cdot|x)} \quad \underbrace{r_\phi(x, y)}_{\text{maximize the reward}} \quad - \underbrace{\beta D_{\text{KL}}\left(\pi(\cdot \mid x) \,\|\, \pi_{\text{ref}}(\cdot \mid x)\right)}_{\text{but don't change the model too much}}$$

- $\pi_{\text{ref}}$ is the reference model, usually the SFT model
- $\beta$ sets the trade-off between improving behavior and staying close to what the model already knows
- This matters because we start from a strong pretrained model, not from a random policy

**Question:** if we could solve this exactly, what would the optimal policy look like?

Based on Lambert, RLHF (rlhfbook.com), and Truong, Haupt, Koyejo, MLHP (mlhp.stanford.edu)

# The optimal policy: set-up

The objective separates across prompts, so fix $x$ and drop it from the notation. We treat responses as a finite set and write $\pi(y)$ for the probabilities:

$$F(\pi) = \sum_y \pi(y)\, r(y) - \beta \sum_y \pi(y) \log \frac{\pi(y)}{\pi_{\text{ref}}(y)} \quad \text{subject to} \quad \sum_y \pi(y) = 1.$$

$F$ is concave in $\pi$, because the first term is linear and negative entropy is convex. So a stationary point of the Lagrangian is a global maximum:

$$\mathcal{L}(\pi, \lambda) = F(\pi) + \lambda \left( \sum_y \pi(y) - 1 \right).$$

# The optimal policy: derivation

Differentiate with respect to one $\pi(y)$ and set the result to zero:

$$r(y) - \beta \left( \log \frac{\pi(y)}{\pi_{\text{ref}}(y)} + 1 \right) + \lambda = 0.$$

Isolate the log-ratio:

$$\log \frac{\pi(y)}{\pi_{\text{ref}}(y)} = \frac{r(y)}{\beta} + \frac{\lambda - \beta}{\beta}.$$

Exponentiate, and write the constant as $e^{(\lambda-\beta)/\beta} = 1/Z$. The constraint $\sum_y \pi(y) = 1$ then fixes $Z$:

$$\pi(y) = \frac{1}{Z} \pi_{\text{ref}}(y)\, e^{r(y)/\beta}, \qquad Z = \sum_y \pi_{\text{ref}}(y)\, e^{r(y)/\beta}.$$

# The optimal policy: what it means

$$\pi^*(y \mid x) = \frac{\pi_{\text{ref}}(y \mid x)\, e^{r(x,y)/\beta}}{Z(x)}$$

The optimal policy **reweights the reference model** exponentially in the reward.

- $\beta \to \infty$: $\pi^* \to \pi_{\text{ref}}$, so nothing changes
- $\beta \to 0$: $\pi^*$ concentrates on the highest-reward responses in the support of $\pi_{\text{ref}}$

**Why not just use this formula?** $Z(x)$ sums over all possible responses, and the reward is itself only an estimate. So we need algorithms that approximate $\pi^*$.

Based on Lambert, RLHF (rlhfbook.com), and Truong, Haupt, Koyejo, MLHP (mlhp.stanford.edu)

# A preview of session 3

Take the log of the optimal policy and solve for the reward:

$$r(x, y) = \beta \log \frac{\pi^*(y \mid x)}{\pi_{\text{ref}}(y \mid x)} + \beta \log Z(x).$$

Now compare two responses to the same prompt. $Z(x)$ doesn't depend on $y$, so it cancels:

$$r(x, y_w) - r(x, y_l) = \beta \log \frac{\pi^*(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \beta \log \frac{\pi^*(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)}.$$

**The intractable constant is gone,** and only log-probabilities of the model remain. This is the key idea of DPO (Rafailov et al., 2023): the sequence log-probabilities from earlier are enough. The implicit reward $\beta \log \frac{\pi_\theta}{\pi_{\text{ref}}}$ plugs straight into Bradley-Terry (Truong et al., 2025).

# Instruction tuning

# Chat templates

A base model only sees a sequence of tokens. A **chat template** turns a list of messages into one flat sequence:

```
<|im_start|>system
You are a friendly chatbot who always responds in the style of a pirate<|im_end|>
<|im_start|>user
How many helicopters can a human eat in one sitting?<|im_end|>
<|im_start|>assistant
```

- **Roles:** `system` holds instructions, `user` holds the person's messages, and `assistant` holds the model's responses
- **Generation cue:** the final `<|im_start|>assistant` tells the model that it's its turn
- **Stopping:** the model generates until it produces the end token, here `<|im_end|>`

Each model family has its own format. Source: Chapter 4.

# What instruction tuning does differently

The loss is the same as pretraining. The differences are in the data and the set-up:

- **Prompt masking:** the model isn’t trained to predict user messages, only responses
- **Smaller batches:** to fit a narrower data distribution while keeping what pretraining taught
- **Lower learning rate:** typically one to two orders of magnitude below pretraining

The main effect is simple: **a base model continues text; an instruction-tuned model answers and stops.**

Learning to stop is a good example of something SFT must teach. The target includes the end token, so the model is trained to produce it.

# What makes instruction tuning work

Principles from Chapter 4:

- **Data quality is key.** The completions are what the model actually learns from.
- **Small can be enough for chat.** Focused datasets can do well in narrow domains (Zhou et al., 2023). For example, No Robots has about 10K human-written samples.
- **Match the use case.** The best prompts resemble the tasks you care about downstream.
- **Later stages can recover from some noise.** The overall recipe matters more than any single stage.

# Lab: from a base model to an assistant

# Lab: what you'll do

**Open the notebook in Colab or Kaggle, and choose a T4 GPU:** Open the lab in Colab. All notebooks are at github.com/pegahani/rlhf-course.

1. **Run the base model** on a few prompts
2. **Instruction-tune it** on a free GPU, using a chat template and a completion mask
3. **Compare before and after:** does the model answer, and does it stop?

Follow the notebook and work in pairs, especially if Colab has no GPU available for you. The exercises have hints and solutions, so use them if you get stuck.

# What we saw

One test run on a free T4, on 32 held-out prompts:

| | Before | After |
|---|---|---|
| **Stopped by itself** | 5/32 | 15/32 |
| **95% interval** | [0.07, 0.32] | [0.31, 0.64] |

**Paired:** 13 prompts stopped only after training, and 3 only before.

How do your numbers compare?

USER

What is the capital of France?

ASSISTANT (After training)

The capital of France is Paris.

USER

Write a haiku about programming.

ASSISTANT (After training)

The best way to learn programming is to do it.

**To discuss:** did it learn to stop, or also learn something new? Is stopping the same as quality?

# Before next session

**Homework, due at session 2:** both notebooks are at github.com/pegahani/rlhf-course

- **HW0, do it first:** from NumPy to PyTorch, practising today's tools

**HW1: the Bradley-Terry model**

- **Foundation:** simulate Bradley-Terry data and fit the model
- **Core:** build confidence intervals and test their coverage
- **Deep:** Fisher information, and coverage when comparisons are correlated

**Reading for session 2**

- **Lambert, *RLHF*:** Chapter 5 (Bradley-Terry reward models), and Chapters 10 and 11
- ***Machine Learning from Human Preferences*:** Chapter 1, especially Sections 1.2, 1.6 and 1.9, and Section 2.3

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